

SDG14: Conserve and sustainably use the oceans, seas and marine resources for sustainable development

SUMMARY

Original search string: 287.783 document results

TITLE-ABS-KEY((marine OR ocean* OR sea OR seas) AND ({water cycle} OR {water cycles} OR {biogeochemical cycle} OR {biogeochemical cycles} OR {biochemical cycle} OR {biochemical cycles} OR {glacier dynamics} OR {glacier mass} OR {glacier retreat} OR {global circulation model} OR {global circulation models} OR {global circulation modelling} OR {global circulation modeling} OR {ice-ocean} OR {submarine geophysics} OR eutrophicat* OR marine OR ocean* OR sea OR seas OR {coral bleach} OR {coral bleaching} OR {coastal management} OR paleoceanography OR paleoceanography OR paleo-oceanography OR {coastal habitat} OR {coastal habitats} OR {marine debris})) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



Updated query (4): 96,446 document results

TITLE-ABS-KEY((marine OR ocean OR oceans OR sea OR seas OR coast* OR mangrove) AND ({water cycle} OR {water cycles} OR {biogeochemical cycle} OR {biogeochemical cycles} OR {oceanic circulation model} OR {oceanic circulation models} OR {oceanic circulation modelling} OR {oceanic circulation modeling} OR {ice-ocean} OR eutrophicat* OR marine OR {coral bleach} OR {coral bleaching} OR {coastal management} OR {coastal habitat} OR {coastal habitats} OR {marine debris} OR {ocean acidification} OR (acidification AND seawater) OR {fishery} OR {fisheries} OR {overfishing})

R {sustainable yield} OR {marine protected area} OR {marine protected areas} OR {marine conservation} OR {ecotourism} OR {community based conservation} OR {community-based conservation} OR {marine land slide} OR {marine pollution} OR {nutrient runoff} OR {coastal ecotourism} OR {destructive fishing} OR {local fisheries} OR {artisanal fishers} OR {fisheries rights} OR {species richness} OR {traditional ecological knowledge} OR {small Island development states} OR {marine quota} OR {marine economy} OR {marine policy}) AND NOT ({paleoclimate} OR {paleoceanography} OR {radiocarbon} OR {genetics} OR {medicine} OR {drug} OR {engineering} OR {aerosol})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

PROCESS

The first stage of this review was to compare the original search string to the targets listed under the Sustainable Development Goal (SDG).

[SDG targets listed here: https://sustainabledevelopment.un.org/sdg14](https://sustainabledevelopment.un.org/sdg14)

The purpose of this stage was to understand whether the Scopus results correctly reflected the theme of the SDG. The SDGs were unanimously agreed by member states of the United Nations, with very deliberate language and scope/ exclusions, so the results of this analysis should accurately reflect the specific SDG themes.

The SDG promotes the **conservation and sustainable use of marine (oceans, seas) and coastal ecosystems and resources**. Effects of **acidification** and **eutrophication** are aimed to be minimised, particularly in relation to **marine debris** and **nutrient pollution** induced by land-based activities. **Sustainable fishing practices**, sustainable yields, **aquaculture** and **ecotourism** are promoted and overfishing, destructive fishing practices and other disturbance factors should be minimised to protect these ecosystems and to increase their resilience. **Equal opportunities for local fisheries** and the establishment of **marine protected areas** are promoted.

A Scopus search was conducted using the original search string. This returned 287.783 document results.

The top 20 most cited articles were reviewed. 12 of the 20 articles do not closely reflect the theme of the SDG. Specifically, these articles related to topics in the field of medicine or biochemical cycles. The latter lacked a specific focus on marine ecosystems. Articles that are not closely related to the SDG also covered topics related to climate change (hit on sea ice - not closely related to the SDG) and terrestrial issues such as drought or aridity, induced by changing marine processes.

Top 20 most cited articles listed in Scopus from the original search string:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Climate change 2013 the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	3483	2-s2.0-84926664150
Global epidemiology of hepatitis C virus infection: New estimates of age-specific antibody to HCV seroprevalence	2013	Hepatology	1573	2-s2.0-84876149257
Global, regional, and national incidence, prevalence, and years lived with disability for 310 diseases and injuries, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015	2016	The Lancet	1471	2-s2.0-84994092049
Increasing drought under global warming in observations and models	2013	Nature Climate Change	1263	2-s2.0-84871840877
Anthropogenic and natural radiative forcing	2013	Climate Change 2013 the Physical Science Basis: Working Group I	1261	2-s2.0-84928644022

		Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change		
Plastic waste inputs from land into the ocean	2015	Science	1233	2-s2.0-84923095173
Noaa's hysplit atmospheric transport and dispersion modeling system	2015	Bulletin of the American Meteorological Society	1069	2-s2.0-84940823379
Scaling, self-similarity, and intermediate asymptotics	2014	Scaling, Self-Similarity, and Intermediate Asymptotics	1047	2-s2.0-84953280783
Global estimates of cancer prevalence for 27 sites in the adult population in 2008	2013	International Journal of Cancer	1044	2-s2.0-84872853140
Natural products: A continuing source of novel drug leads	2013	Biochimica et Biophysica Acta - General Subjects	998	2-s2.0-84876002671
Carbon and other biogeochemical cycles	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	913	2-s2.0-84928642948
Biochemical adaptation	2014	Biochemical Adaptation	907	2-s2.0-85022191771
Status and ecological effects of the world's largest carnivores	2014	Science	905	2-s2.0-84892185014
The NCEP climate forecast system version 2	2014	Journal of Climate	895	2-s2.0-84896093136
Impacts of ocean acidification on marine organisms: Quantifying sensitivities and interaction with warming	2013	Global Change Biology	824	2-s2.0-84877016612
Bedmap2: Improved ice bed, surface and thickness datasets for Antarctica	2013	Cryosphere	763	2-s2.0-84874462170
Plastic Pollution in the World's Oceans: More than 5 Trillion Plastic Pieces Weighing over 250,000 Tons Afloat at Sea	2014	PLoS ONE	758	2-s2.0-84914176492
Recent global-warming hiatus tied to equatorial Pacific surface cooling	2013	Nature	756	2-s2.0-84884410748
Global, regional, and national age-sex specific mortality for 264 causes of death, 1980-2016: A systematic analysis for the Global Burden of Disease Study 2016	2017	The Lancet	733	2-s2.0-85031727040
The physical impacts of microplastics on marine organisms: a review.	2013	Environmental pollution (Barking, Essex : 1987)	732	2-s2.0-84970014901

In addition, a randomised sample of 20 articles were reviewed. 14 of the 20 articles do not closely reflect the theme of the SDG. Specifically, these articles related to the fields of earth science, archaeology. These were retrieved through hits on the keyword 'palaeoceanography'. Other articles related to biochemistry, engineering and biological sciences in general.

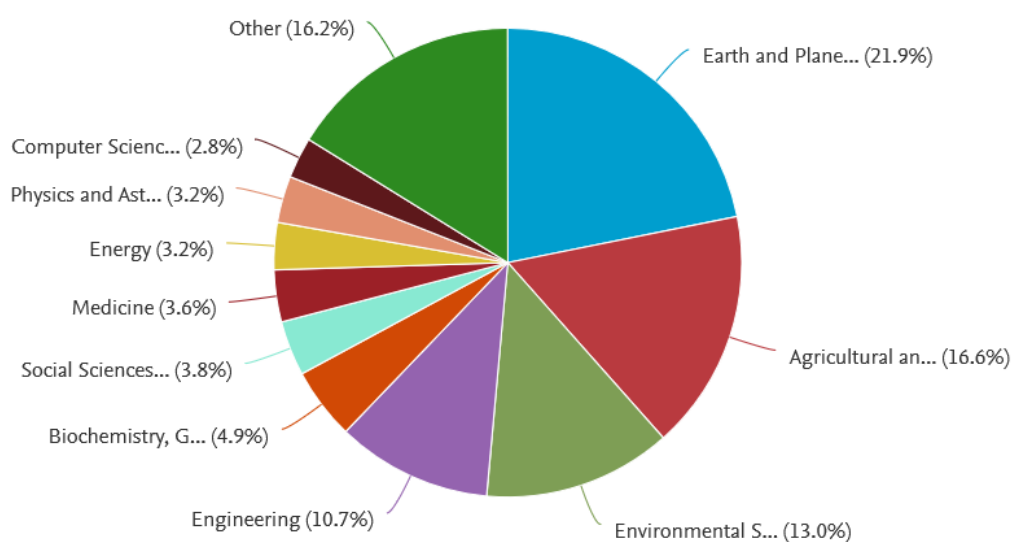
Randomised sample of 20 articles listed in Scopus from the original search string:

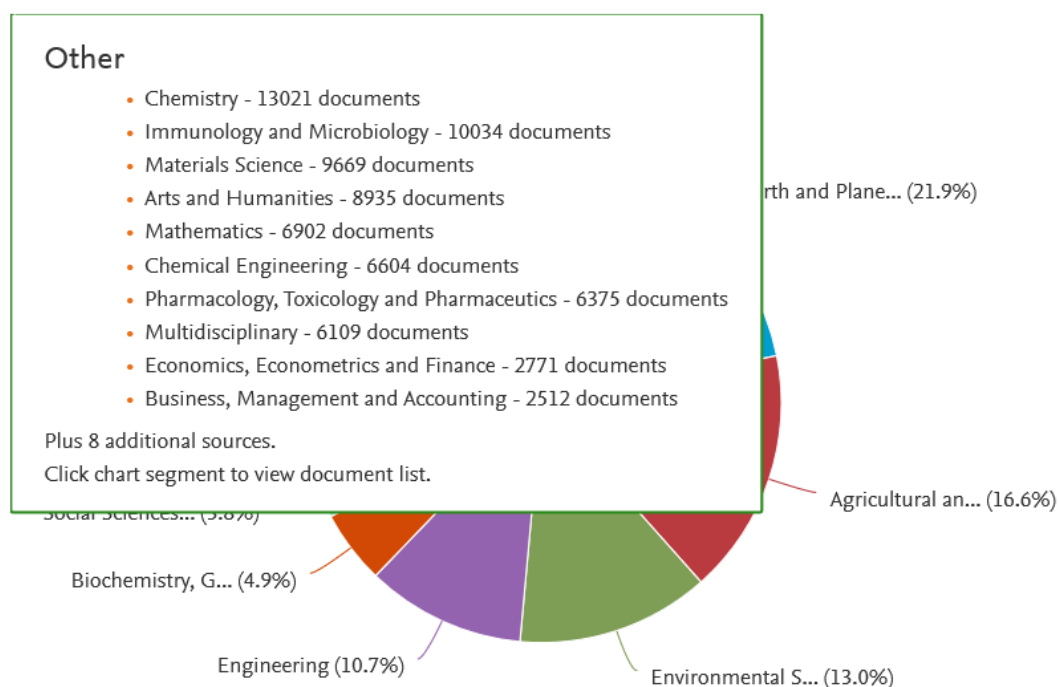
*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Canopy flow analysis reveals the advantage of size in the oldest communities of multicellular eukaryotes	2014	Current Biology	37	2-s2.0-84895068488
Characterization of the mantle transcriptome of Yesso scallop (<i>Patinopecten yessoensis</i>): Identification of genes potentially involved in biomineralization and pigmentation	2015	PLoS ONE	35	2-s2.0-84928914609
A new genus and new species of Desmosomatidae Sars, 1897 (Isopoda) from the eastern South Atlantic abyss described by means of integrative taxonomy	2015	Marine Biodiversity	24	2-s2.0-84939878380
'Anchialine' redefined as a subterranean estuary in a crevicular or cavernous geological setting	2015	Journal of Crustacean Biology	20	2-s2.0-84938073236
Role of crustal contribution in the early stage of the Damara Orogen, Namibia: New constraints from combined U-Pb and Lu-Hf isotopes from the Goas Magmatic Complex	2015	Gondwana Research	18	2-s2.0-84930235847
MIS 13-12 in Britain and the North Atlantic: Understanding the palaeoclimatic context of the earliest Acheulean	2015	Journal of Quaternary Science	12	2-s2.0-84955189038
Phylogeny and systematics of mitriform gastropods (Mollusca: Gastropoda: Neogastropoda)	2015	Zoological Journal of the Linnean Society	9	2-s2.0-84942190877
Distribution of radionuclides in surface seawater obtained by an aerial radiological survey	2014	Journal of Nuclear Science and Technology	7	2-s2.0-84903545920
Changes in area-level socioeconomic status and oral health of indigenous Australian children	2016	Journal of Health Care for the Poor and Underserved	4	2-s2.0-84957959786
Amylase activity of aquatic actinomycetes isolated from the sediments of mangrove forests in south of Iran	2015	Egyptian Journal of Aquatic Research	4	2-s2.0-84930575989
First geolocator tracks of Swedish red-necked phalaropes reveal the Scandinavia-Arabian Sea connection	2016	Journal of Avian Biology	3	2-s2.0-84949293967
Ocean Surface Geostrophic Circulation Climatology and Annual Variations Inferred from Satellite Altimetry and GOCE Gravity Data	2016	Pure and Applied Geophysics	2	2-s2.0-84961285146
Penidioxolanes A and B, 1,3-dioxolane containing azaphilone derivatives from marine-derived <i>Penicillium</i> sp. KCB12C078	2015	Natural Product Sciences	2	2-s2.0-84957881179
Unlocking the potential of remaining hydrocarbons using time lapse pulsed neutron capture logs in complex completions - A case study from UK Central North Sea	2016	Society of Petroleum Engineers - SPE Europec Featured	1	2-s2.0-84992156957

		at 78th EAGE Conference and Exhibition		
How warming oceans unleashed an ice stream	2015	Science	1	2-s2.0- 84946909464
Prehistoric walrus hunting site shielded	2017	Science	0	2-s2.0- 85010297634
UNDERWATER REPAIRS TO EKOFISK STRUCTURES.	2017	Proceedings of the Annual Offshore Technology Conference	0	2-s2.0- 0016867279
Master thesis students in the video game industry	2015	SIGGRAPH Asia 2015 RDVG in the Video Game Industry, SA 2015	0	2-s2.0- 84959096651
Discuss on the two algorithms of line-segments and dot-array for region judgement of the sub-satellite purview	2015	Proceedings of SPIE - The International Society for Optical Engineering	0	2-s2.0- 84947724254
Proposed rules on endangered species and clean water: The impacts on project development	2015	IEEE Cement Industry Technical Conference (Paper)	0	2-s2.0- 84945242995

Documents by subject area





The results of the analyses are confirmed by the general analysis of the overall documents results. A large number of articles are retrieved from the fields of 'Earth and Planetary Sciences', 'Agricultural and Biological Sciences' and 'Environmental Sciences'. A number of the articles that were classified as 'not closely related to the theme of the SDG' are part of these fields, e.g. articles that cover topics on evolution, fossils, paleo-sciences or climate change. The general analysis shows that a large proportion of articles are retrieved from the field of 'Engineering', as found in the randomised sample. The 'Other' section contains a large number of irrelevant articles in relation to 'Chemistry' and 'Immunology and Microbiology'. The co-occurrence of keywords confirms these findings and shows a better focus is needed on topics / keywords that specifically relate to the targets of the SDG. This search query could be improved, and a first step will be to either remove or narrow down existing keywords.

Update 1) The search keywords were edited to remove the broader search terms, or to make them more specific by combining with other search keywords.

Original search string: 287.783 document results

TITLE-ABS-KEY((marine OR ocean* OR sea OR seas) AND ({water cycle} OR {water cycles} OR {biogeochemical cycle} OR {biogeochemical cycles} OR {biochemical cycle} OR {biochemical cycles} OR {glacier dynamics} OR {glacier mass} OR {glacier retreat} OR {global circulation model} OR {global circulation models} OR {global circulation modelling} OR {global circulation modeling} OR {ice-ocean} OR {submarine geophysics} OR eutrophicat* OR marine OR ocean* OR sea OR seas OR {coral bleach} OR {coral bleaching} OR {coastal management} OR paleoceanography OR paleoceanography OR paleo-oceanography OR {coastal habitat} OR {coastal habitats} OR {marine debris})) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Rationale for edits to the search keywords:

<u>query term</u>	<u>decision</u>	<u>rationale</u>
{Glacier dynamics}	Deleted	Results in hits on glacier related topics, which would be a better match with SDG 13 (climate change) than SDG14 on sustainable use of

		marine resources and protection of marine resources
{Glacier mass}	Deleted	Results in hits on glacier related topics, which would be a better match with SDG 13 (climate change) than SDG14 on sustainable use of marine resources and protection of marine resources
{Glacier retreat}	Deleted	Results in hits on glacier related topics, which would be a better match with SDG 13 (climate change) than SDG14 on sustainable use of marine resources and protection of marine resources
paleoceanography	Deleted	A field that focusses on the history of climate change or the earth's environment (fossils etc), indeed by conducting research in a marine environment, but does not match the theme of SDG14
paleoceanograph y	Deleted	A field that focusses on the history of climate change or the earth's environment (fossils etc), indeed by conducting research in a marine environment, but does not match the theme of SDG14
paleo- oceanography	Deleted	A field that focusses on the history of climate change or the earth's environment (fossils etc), indeed by conducting research in a marine environment, but does not match the theme of SDG14
Ocean*	Changed to two key words: ocean OR oceans	Ocean* led to hits on articles that are irrelevant to the theme of SDG 14, which are conducted or applied to areas in Oceania.
global circulation model	Changed to {oceanic circulation model}	Global circulation model is too broad as these also include atmospheric models, narrowed down to oceanic circulation model
global circulation models	Changed to {oceanic circulation models}	Global circulation models are too broad as these also include atmospheric models, narrowed down to oceanic circulation models
global circulation modelling	Changed to {oceanic circulation modelling}	Global circulation modelling is too broad as this also includes atmospheric modelling, narrowed down to oceanic circulation modelling
global circulation modeling	Changed to {oceanic circulation modelling}	Global circulation modelling is too broad as this also includes atmospheric modelling, narrowed down to oceanic circulation modelling
Biochemical cycle	Deleted	Part of biogeochemical cycle, but this keyword could also lead to results on internal biological cycles and the cycle between biotic (animal / plant individual) and abiotic environment which is not the purpose of this SDG
Biochemical cycles	Deleted	Part of biogeochemical cycle, but this keyword could also lead to results on internal biological cycles and the cycle between biotic (animal / plant individual) and abiotic environment which is not the purpose of this SDG
Ocean*	Deleted	Deleted as these are already in the first equation of the query. It was found that this led to less hits on irrelevant topics such as engineering, terrestrial studies which mention oceans, etc; marine as additional indicator is sufficient to identify articles that cover topics related to a marine environment or marine species.

sea	Deleted	Deleted as these are already in the first equation of the query. It was found that this led to less hits on irrelevant topics such as engineering, terrestrial studies which mention oceans, etc; marine as additional indicator is sufficient to identify articles that cover topics related to a marine environment or marine species.
Seas	Deleted	Deleted as these are already in the first equation of the query. It was found that this led to less hits on irrelevant topics such as engineering, terrestrial studies which mention oceans, etc; marine as additional indicator is sufficient to identify articles that cover topics related to a marine environment or marine species.

Updated search string (first update): 112,691 document results

TITLE-ABS-KEY((marine OR ocean OR oceans OR sea OR seas) AND ({water cycle} OR {water cycles} OR {biogeochemical cycle} OR {biogeochemical cycles} OR {oceanic circulation model} OR {oceanic circulation models} OR {oceanic circulation modelling} OR {oceanic circulation modeling} OR {ice-ocean} OR {submarine geophysics} OR eutrophicat* OR marine OR {coral bleach} OR {coral bleaching} OR {coastal management} OR {coastal habitat} OR {coastal habitats} OR {marine debris})) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search returned 112,691 document results, compared to document results for the original search.

The top 20 cited articles were reviewed. Six of the 20 articles do not closely reflect the theme of the SDG. Specifically, these articles related to toxicity of chemicals, historical records of climate change and sedimentation, and atmospheric pollution.

Top 20 most cited articles listed in Scopus from the search string (v1):

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Climate change 2013 the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	3483	2-s2.0-84926664150
Plastic waste inputs from land into the ocean	2015	Science	1233	2-s2.0-84923095173
Natural products: A continuing source of novel drug leads	2013	Biochimica et Biophysica Acta - General Subjects	998	2-s2.0-84876002671
Carbon and other biogeochemical cycles	2013	Climate Change 2013 the Physical	913	2-s2.0-84928642948

		Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change		
Impacts of ocean acidification on marine organisms: Quantifying sensitivities and interaction with warming	2013	Global Change Biology	824	2-s2.0-84877016612
Plastic Pollution in the World's Oceans: More than 5 Trillion Plastic Pieces Weighing over 250,000 Tons Afloat at Sea	2014	PLoS ONE	758	2-s2.0-84914176492
The physical impacts of microplastics on marine organisms: a review.	2013	Environmental pollution (Barking, Essex : 1987)	732	2-s2.0-84970014901
Toxicity, mechanism and health effects of some heavy metals	2014	Interdisciplinary Toxicology	638	2-s2.0-84910613290
The Collection 6 MODIS aerosol products over land and ocean	2013	Atmospheric Measurement Techniques	635	2-s2.0-84887281897
Mercury as a global pollutant: Sources, pathways, and effects	2013	Environmental Science and Technology	615	2-s2.0-84878224971
Recent and planned developments of the program oxcal	2013	Radiocarbon	611	2-s2.0-84884408062
Global imprint of climate change on marine life	2013	Nature Climate Change	599	2-s2.0-84886722508
Processes and patterns of oceanic nutrient limitation	2013	Nature Geoscience	586	2-s2.0-84883444382
Global conservation outcomes depend on marine protected areas with five key features	2014	Nature	572	2-s2.0-84893920384
Global warming and recurrent mass bleaching of corals	2017	Nature	552	2-s2.0-85015676975
Renewable energy resources: Current status, future prospects and their enabling technology	2014	Renewable and Sustainable Energy Reviews	511	2-s2.0-84905676737
Microplastic ingestion by zooplankton	2013	Environmental Science and Technology	501	2-s2.0-84879246992
A stratigraphic framework for abrupt climatic changes during the Last Glacial period based on three synchronized Greenland ice-core records: Refining and extending the INTIMATE event stratigraphy	2014	Quaternary Science Reviews	492	2-s2.0-84924351306
Contribution of Antarctica to past and future sea-level rise	2016	Nature	487	2-s2.0-84962381525
Extended reconstructed sea surface temperature version 4 (ERSST.v4). Part I: Upgrades and intercomparisons	2015	Journal of Climate	485	2-s2.0-84923008551

In addition, a randomised sample of 20 articles was reviewed. 15 of 20 articles do not closely reflect the theme of the SDG. Several articles covered topics on historical records of climate change or

atmospheric pollution, the majority covered topics related to the field of engineering and (bio)chemistry.

Randomised sample of 20 articles listed in Scopus from the search string (v1):

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Climate model and proxy data constraints on ocean warming across the Paleocene-Eocene Thermal Maximum	2013	Earth-Science Reviews	106	2-s2.0-84882751221
A cross-scale numerical modeling system for management support of oil spill accidents	2014	Marine Pollution Bulletin	35	2-s2.0-84896401915
Sensitivity analysis of the use of Life Cycle Impact Assessment methods: A case study on building materials	2016	Journal of Cleaner Production	24	2-s2.0-84958102653
Durability performance and service life of CFCC tendons exposed to elevated temperature and alkaline environment	2016	Journal of Composites for Construction	19	2-s2.0-84955604824
Targeting cancer with sesterterpenoids: The new potential antitumor drugs	2015	Journal of Natural Medicines	18	2-s2.0-84930762344
Genome sequencing reveals metabolic and cellular interdependence in an amoeba-kinetoplastid symbiosis	2017	Scientific Reports	13	2-s2.0-85029498112
Characterization of carbohydrates in rainwater from the Southeastern North Carolina	2014	Chemosphere	12	2-s2.0-84901441830
Maintenance grouping optimization for the management of risk in offshore riser system	2015	Process Safety and Environmental Protection	10	2-s2.0-84948176881
Microstructure and Corrosion Behaviour of Laser Metal Deposited Ti6Al4V/Cu Composites in 3.5% Sea Water	2015	Materials Today: Proceedings	9	2-s2.0-84947306761
Influence of Sea Water Aging on the Mechanical Behaviour of Acrylic Matrix Composites	2017	Applied Composite Materials	8	2-s2.0-84979671301
The Lilliput effect in crinoids at the end of the Oceanic Anoxic Event 2: A Case study from Poland	2015	Journal of Paleontology	8	2-s2.0-84971457383
From the peat bog to the estuarine mixing zone: Common features and variances in riverine dissolved organic matter determined by non-targeted analysis	2017	Marine Chemistry	6	2-s2.0-85023616338
Sounding the aquapelago: The cultural-environmental context of ni-Vanuatu women's liquid percussion performance	2014	Perfect Beat	6	2-s2.0-84941638016
Local flow characterization using bioinspired sensory information	2017	Journal of Fluid Mechanics	4	2-s2.0-85016578839
Efficacy of baker's yeast as immunostimulant in Nile tilapia (<i>Oreochromis niloticus</i>)	2015	International Journal of ChemTech Research	2	2-s2.0-84935865636
Tests for temperature and pressure response of silt soft soil under blasting load	2014	Zhendong yu Chongji/Journal of Vibration and Shock	1	2-s2.0-84920521648
Duplexer based on a cavity resonators for PSR	2016	Proceedings Elmar - International Symposium	0	2-s2.0-84996939047

The new search results were compared to the original search results to check if the updated search was primarily removing irrelevant results.

The original and updated search queries were combined using the AND NOT function in Scopus. This returned 175,092 document results – i.e. the publications from the first search that were excluded in the second search.

The top 20 most cited articles were reviewed. Eighteen of the 20 articles do not closely reflect the theme of the SDG. The majority of these articles cover topics related to climate change projections, terrestrial ecosystems, atmospheric cycles and the field of medicine.

Top 20 cited articles included in the original search results but not in the updated search results:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Global epidemiology of hepatitis C virus infection: New estimates of age-specific antibody to HCV seroprevalence	2013	Hepatology	1573	2-s2.0-84876149257
Global, regional, and national incidence, prevalence, and years lived with disability for 310 diseases and injuries, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015	2016	The Lancet	1471	2-s2.0-84994092049
Increasing drought under global warming in observations and models	2013	Nature Climate Change	1263	2-s2.0-84871840877
Anthropogenic and natural radiative forcing	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1261	2-s2.0-84928644022
Noaa's hysplit atmospheric transport and dispersion modeling system	2015	Bulletin of the American Meteorological Society	1069	2-s2.0-84940823379
Scaling, self-similarity, and intermediate asymptotics	2014	Scaling, Self-Similarity, and Intermediate Asymptotics	1047	2-s2.0-84953280783
Global estimates of cancer prevalence for 27 sites in the adult population in 2008	2013	International Journal of Cancer	1044	2-s2.0-84872853140
Biochemical adaptation	2014	Biochemical Adaptation	907	2-s2.0-85022191771
Status and ecological effects of the world's largest carnivores	2014	Science	905	2-s2.0-84892185014
The NCEP climate forecast system version 2	2014	Journal of Climate	895	2-s2.0-84896093136
Bedmap2: Improved ice bed, surface and thickness datasets for Antarctica	2013	Cryosphere	763	2-s2.0-84874462170
Recent global-warming hiatus tied to equatorial Pacific surface cooling	2013	Nature	756	2-s2.0-84884410748
Global, regional, and national age-sex specific mortality for 264 causes of death,	2017	The Lancet	733	2-s2.0-85031727040

1980-2016: A systematic analysis for the Global Burden of Disease Study 2016				
Global prevalence of age-related macular degeneration and disease burden projection for 2020 and 2040: A systematic review and meta-analysis	2014	The Lancet Global Health	729	2-s2.0-84892965801
The rise of oxygen in Earth's early ocean and atmosphere	2014	Nature	707	2-s2.0-84894261996
Climate change projections using the IPSL-CM5 Earth System Model: From CMIP3 to CMIP5	2013	Climate Dynamics	707	2-s2.0-84876880770
Climate change and the permafrost carbon feedback	2015	Nature	699	2-s2.0-84927514015
The global precipitation measurement mission	2014	Bulletin of the American Meteorological Society	688	2-s2.0-84897665733
Climate change impacts on global food security	2013	Science	654	2-s2.0-84881061285
Gondwana dispersion and Asian accretion: Tectonic and palaeogeographic evolution of eastern Tethys	2013	Journal of Asian Earth Sciences	648	2-s2.0-84875367305

The analyses show that further improvements can be made. Despite the removal of keywords related to palaeoceanography and climate change, respective topics are still retrieved by the updated query term. The co-occurrence of the terms also shows that a large number of keywords, and therefore topics, relate to genetics and engineering. An additional update is needed to exclude these fields and to ensure that the search query retrieves articles that mainly relate to the theme of the SDG.

Update 2) A second update was made to the search string to filter out fields that do not relate to the theme of the SDG.

The objective of the second update was to filter out irrelevant articles related to historical records of climate change, medicine and engineering. The following query terms were added to the search query:

AND NOT

{paleoclimate} OR {paleoceanography} OR {radiocarbon}; to remove any articles related to historical climate change studies

OR {genetics} OR {medicine} OR {drug}; to remove health-care related articles, often focussing on medicine

OR {engineering} ; to remove any articles mainly focussing on engineering

OR {aerosol} ; to remove any articles related to atmospheric pollution and aerosols specifically

Second update search: 86,237 document results

TITLE-ABS-KEY ((marine OR ocean OR oceans OR sea OR seas) AND ({water cycle} OR {water cycles} OR {biogeochemical cycle} OR {biogeochemical cycles} OR {oceanic circulation model} OR {oceanic circulation models} OR {oceanic circulation modelling} OR {oceanic circulation modeling} OR {ice-ocean} OR {submarine geophysics} OR eutrophicat* OR marine OR {coral bleach} OR {coral bleaching} OR {coastal management} OR {coastal habitat} OR {coastal habitats} OR {marine debris}) AND NOT ({paleoclimate} OR {paleoceanography} OR {radiocarbon} OR {genetics} OR {medicine} OR {drug} OR {engineering} OR {aerosol})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The top 20 cited articles were reviewed. Three of the 20 articles do not closely reflect the theme of the SDG. Specifically, these articles related to biochemistry and renewable resources.

Top 20 most cited articles listed in Scopus from the updated (v2) string:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Climate change 2013 the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	3500	2-s2.0-84926664150
Plastic waste inputs from land into the ocean	2015	Science	1239	2-s2.0-84923095173
Carbon and other biogeochemical cycles	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	915	2-s2.0-84928642948
Impacts of ocean acidification on marine organisms: Quantifying sensitivities and interaction with warming	2013	Global Change Biology	826	2-s2.0-84877016612
Plastic Pollution in the World's Oceans: More than 5 Trillion Plastic Pieces Weighing over 250,000 Tons Afloat at Sea	2014	PLoS ONE	766	2-s2.0-84914176492
Toxicity, mechanism and health effects of some heavy metals	2014	Interdisciplinary Toxicology	644	2-s2.0-84910613290
Mercury as a global pollutant: Sources, pathways, and effects	2013	Environmental Science and Technology	617	2-s2.0-84878224971
Global imprint of climate change on marine life	2013	Nature Climate Change	601	2-s2.0-84886722508
Processes and patterns of oceanic nutrient limitation	2013	Nature Geoscience	589	2-s2.0-84883444382
Global conservation outcomes depend on marine protected areas with five key features	2014	Nature	572	2-s2.0-84893920384
Global warming and recurrent mass bleaching of corals	2017	Nature	556	2-s2.0-85015676975
Renewable energy resources: Current status, future prospects and their enabling technology	2014	Renewable and Sustainable Energy Reviews	513	2-s2.0-84905676737

Microplastic ingestion by zooplankton	2013	Environmental Science and Technology	504	2-s2.0-84879246992
Contribution of Antarctica to past and future sea-level rise	2016	Nature	487	2-s2.0-84962381525
Extended reconstructed sea surface temperature version 4 (ERSST.v4). Part I: Upgrades and intercomparisons	2015	Journal of Climate	485	2-s2.0-84923008551
Multiple stressors of ocean ecosystems in the 21st century: Projections with CMIP5 models	2013	Biogeosciences	463	2-s2.0-84885110424
Evaluation of the ECMWF ocean reanalysis system ORAS4	2013	Quarterly Journal of the Royal Meteorological Society	454	2-s2.0-84880703032
Biodiesel from microalgae: A critical evaluation from laboratory to large scale production	2013	Applied Energy	453	2-s2.0-84871717568
Extinction risk and conservation of the world's sharks and rays	2014	eLife	450	2-s2.0-84898714955
Marine ecological processes	2016	Marine Ecological Processes	449	2-s2.0-84974777289

In addition, a randomised sample of 20 articles was reviewed. Eight of the 20 articles do not closely reflect the theme of the SDG. It is noted that highly irrelevant articles are now limited to three articles, these articles covered topics related to engineering and terrestrial ecosystems. Five of the 20 articles relate to historical marine ecosystems and marine fossils; these topics are not considered to be closely related to the theme of the SDG.

Sample of 20 randomised articles listed in Scopus from the updated (v2) string:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Quiet(er) marine protected areas	2015	Marine Pollution Bulletin	32	2-s2.0-84941771909
Characterizing the distribution of methane sources and cycling in the deep sea via in situ stable isotope analysis	2013	Environmental Science and Technology	25	2-s2.0-84873421979
Thermal erosion of cratonic lithosphere as a potential trigger for mass-extinction	2016	Scientific Reports	18	2-s2.0-84962233182
Quantifying the human influence on fire ignition across the western USA	2016	Ecological Applications	15	2-s2.0-85000916407
The jellyfish buffet: Jellyfish enhance seabird foraging opportunities by concentrating prey	2015	Biology Letters	14	2-s2.0-84946606222
On smoke rings produced by a loop buried in a conductive half-space	2015	Geophysics	11	2-s2.0-84935834264
Entwisleia bella, gen. et sp. nov., a novel marine 'batrachospermaceous' red alga from southeastern Tasmania representing a new family and order in the Nemaliophycidae	2013	European Journal of Phycology	9	2-s2.0-84890604262
Leaching of Metals and Metalloids from Hydrothermal Ore Particulates and Their Effects on Marine Phytoplankton	2017	ACS Omega	8	2-s2.0-85028920722

Corrosion behaviour of chrome-manganese austenitic stainless steels and AISI 304 stainless steel in chloride environment	2014	Transactions of the Indian Institute of Metals	8	2-s2.0-84897859540
The role of marine protected areas and sanctuaries	2014	Sharks: Conservation, Governance and Management	8	2-s2.0-84952941603
The role of external carbonic anhydrase in photosynthesis during growth of the marine diatom <i>Chaetoceros muelleri</i>	2017	Journal of Phycology	6	2-s2.0-85037056157
Shape, colour plasticity, and habitat use indicate morph-specific camouflage strategies in a marine shrimp	2016	BMC Evolutionary Biology	6	2-s2.0-84992188011
CDOM sources and photobleaching control quantum yields for oceanic DMS photolysis	2016	Environmental Science and Technology	4	2-s2.0-85021852471
An analytical framework for understanding tropical meridional modes	2017	Journal of Climate	2	2-s2.0-85017515621
Criteria for assessing maturity of skulls in the common dolphin, <i>Delphinus sp.</i> , from New Zealand waters	2015	Marine Mammal Science	2	2-s2.0-84937022556
Effect of short-term elevated nutrients and mesoherbivore grazing on photosynthesis of macroalgal communities [Eutrofeerumise ja herbivooride lühiajaline mõju makrovetikate koosluste fotosünteesile]	2014	Proceedings of the Estonian Academy of Sciences	2	2-s2.0-84896040490
West coast north America record of the paleogene marine stromboid gastropod <i>rimella</i> and paleobiogeography of the genus	2013	Journal of Paleontology	2	2-s2.0-84884503715
Paleoenvironmental reconstruction of quaternary sedimentation in eastern Sri Lanka: An ichnological study	2017	Journal of the Geological Society of India	1	2-s2.0-85010670236
Biomedical Applications of Enzymes From Marine Actinobacteria	2017	Advances in Food and Nutrition Research	1	2-s2.0-85008234757
Southwest Pacific sea surface conditions during Marine Isotope Stage 11 - Results from dinoflagellate cysts	2016	Palaeogeography, Palaeoclimatology, Palaeoecology	1	2-s2.0-84960928982

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



Combining the second updated search with the original search using AND NOT function in Scopus resulted in 26,470 document results – i.e. the results that were excluded from the original search in the new query.

Marine omega-3 fatty acids and inflammatory processes: Effects, mechanisms and clinical relevance	2015	Biochimica et Biophysica Acta - Molecular and Cell Biology of Lipids	469	2-s2.0-84922553466
Occurrence of microplastics in the gastrointestinal tract of pelagic and demersal fish from the English Channel	2013	Marine Pollution Bulletin	450	2-s2.0-84874713435
Optical methods for sensing and imaging oxygen: Materials, spectroscopies and applications	2014	Chemical Society Reviews	440	2-s2.0-84899094220
Moth-flame optimization algorithm: A novel nature-inspired heuristic paradigm	2015	Knowledge-Based Systems	439	2-s2.0-84944327605
Ingested plastic transfers hazardous chemicals to fish and induces hepatic stress	2013	Scientific Reports	392	2-s2.0-84888340608
Chitin and chitosan preparation from marine sources. Structure, properties and applications	2015	Marine Drugs	388	2-s2.0-84925840401
The role of coastal plant communities for climate change mitigation and adaptation	2013	Nature Climate Change	377	2-s2.0-84885198003
Microplastics in freshwater systems: A review of the emerging threats, identification of knowledge gaps and prioritisation of research needs	2015	Water Research	371	2-s2.0-84925624373
Microplastics in bivalves cultured for human consumption	2014	Environmental Pollution	361	2-s2.0-84903954126
The present and future of microplastic pollution in the marine environment	2014	Environmental Pollution	360	2-s2.0-84891482427
Astaxanthin: Sources, extraction, stability, biological activities and its commercial applications - A review	2014	Marine Drugs	360	2-s2.0-84892855353
Large contribution of natural aerosols to uncertainty in indirect forcing	2013	Nature	348	2-s2.0-84887332636
Trophic level transfer of microplastic: Mytilus edulis (L.) to Carcinus maenas (L.)	2013	Environmental Pollution	333	2-s2.0-84874684591
Chitin and chitosan in selected biomedical applications	2014	Progress in Polymer Science	330	2-s2.0-84906270784

14 of the 20 articles are rightfully excluded. Seven articles were excluded that are closely matching the SDG theme. Taking a closer look at why these seven articles are excluded, it shows that the exclusion of the keyword 'drug' leads to the exclusion of articles on microplastics, as articles on microplastics frequently use keywords such as 'drug effect' or 'unclassified drug'. The exclusion of 'genetics' to avoid the inclusion of articles in the field of medicine or microbiology led to the exclusion of one article.

The most important next step will be to find a way to include articles on microplastics without including articles on medicine / drugs. In addition, further improvements are required to prevent the retrieval of articles on palaeoceanography and fossils.

Update 3) A third update was made by adding keywords that cover topics that were missing from the original and updated queries.

The original, first and second update of the search string contained keywords that primarily retrieve articles related to two targets of the SDG: marine pollution (plastics / nutrients) / debris and coral bleaching.

This means that other targets, related to sustainable fishing practices, sustainable yields, aquaculture, ecotourism, acidification, marine protected areas and equal chances for local fishermen are not part of the search query yet.

The table below lists the different targets of the SDG and the keywords that could potentially help retrieve articles that cover the respective target.

Target	Keywords (to be added)
By 2025, prevent and significantly reduce marine pollution of all kinds, in particular from land-based activities, including marine debris and nutrient pollution	Included
By 2020, sustainably manage and protect marine and coastal ecosystems to avoid significant adverse impacts, including by strengthening their resilience, and take action for their restoration in order to achieve healthy and productive oceans	Included
Minimize and address the impacts of ocean acidification , including through enhanced scientific cooperation at all levels	{acidification}
By 2020, effectively regulate harvesting and end overfishing, illegal, unreported and unregulated fishing and destructive fishing practices and implement science-based management plans, in order to restore fish stocks in the shortest time feasible, at least to levels that can produce maximum sustainable yield as determined by their biological characteristics	{fishery} {fisheries} {overfishing} {sustainable yield}
By 2020, conserve at least 10 per cent of coastal and marine areas , consistent with national and international law and based on the best available scientific information	{marine protected area} {marine protected areas} {marine conservation}
By 2020, prohibit certain forms of fisheries subsidies which contribute to overcapacity and overfishing , eliminate subsidies that contribute to illegal, unreported and unregulated fishing and refrain from introducing new such subsidies, recognizing that appropriate and effective special and differential treatment for developing and least developed countries should be an integral part of the World Trade Organization fisheries subsidies negotiation	See above (fishery, fisheries, overfishing, sustainable yield)

By 2030, increase the economic benefits to Small Island developing States and least developed countries from the sustainable use of marine resources, including through sustainable management of fisheries, aquaculture and tourism	{ecotourism}, {community-based conservation} {community based conservation}
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Third update search query: 92,535 document results

TITLE-ABS-KEY ((marine OR ocean OR oceans OR sea OR seas) AND ({water cycle} OR {water cycles} OR {biogeochemical cycle} OR {biogeochemical cycles} OR {oceanic circulation model} OR {oceanic circulation models} OR {oceanic circulation modelling} OR {oceanic circulation modeling} OR {ice-ocean} OR {submarine geophysics} OR eutrophicat* OR marine OR {coral bleach} OR {coral bleaching} OR {coastal management} OR {coastal habitat} OR {coastal habitats} OR {marine debris} OR {acidification} OR {fishery} OR {fisheries} OR {overfishing} OR {sustainable yield} OR {marine protected area} OR {marine protected areas} OR {marine conservation} OR {ecotourism} OR {community based conservation} OR {community-based conservation}) AND NOT ({paleoclimate} OR {paleoceanography} OR {radiocarbon} OR {genetics} OR {medicine} OR {drug} OR {engineering} OR {aerosol})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search returned 92,535 document results.

The top 20 most cited articles were reviewed. As with search query version 2, three out of 20 articles were irrelevant. Specifically, these articles related to biochemistry and renewable resources.

Top 20 most cited articles listed in Scopus from the updated (v3) string:

***red text shows articles which do not closely match the theme of the SDG**

Title	Year	Source title	Cited by	EID
Climate change 2013 the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	3500	2-s2.0-84926664150
Plastic waste inputs from land into the ocean	2015	Science	1239	2-s2.0-84923095173
Carbon and other biogeochemical cycles	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental	915	2-s2.0-84928642948

		Panel on Climate Change		
Impacts of ocean acidification on marine organisms: Quantifying sensitivities and interaction with warming	2013	Global Change Biology	826	2-s2.0-84877016612
Plastic Pollution in the World's Oceans: More than 5 Trillion Plastic Pieces Weighing over 250,000 Tons Afloat at Sea	2014	PLoS ONE	766	2-s2.0-84914176492
Toxicity, mechanism and health effects of some heavy metals	2014	Interdisciplinary Toxicology	644	2-s2.0-84910613290
Mercury as a global pollutant: Sources, pathways, and effects	2013	Environmental Science and Technology	617	2-s2.0-84878224971
Global imprint of climate change on marine life	2013	Nature Climate Change	601	2-s2.0-84886722508
Processes and patterns of oceanic nutrient limitation	2013	Nature Geoscience	589	2-s2.0-84883444382
Global conservation outcomes depend on marine protected areas with five key features	2014	Nature	572	2-s2.0-84893920384
Global warming and recurrent mass bleaching of corals	2017	Nature	556	2-s2.0-85015676975
Renewable energy resources: Current status, future prospects and their enabling technology	2014	Renewable and Sustainable Energy Reviews	513	2-s2.0-84905676737
Microplastic ingestion by zooplankton	2013	Environmental Science and Technology	504	2-s2.0-84879246992
Contribution of Antarctica to past and future sea-level rise	2016	Nature	487	2-s2.0-84962381525
Extended reconstructed sea surface temperature version 4 (ERSST.v4). Part I: Upgrades and intercomparisons	2015	Journal of Climate	485	2-s2.0-84923008551
Multiple stressors of ocean ecosystems in the 21st century: Projections with CMIP5 models	2013	Biogeosciences	463	2-s2.0-84885110424
Evaluation of the ECMWF ocean reanalysis system ORAS4	2013	Quarterly Journal of the Royal Meteorological Society	454	2-s2.0-84880703032
Biodiesel from microalgae: A critical evaluation from laboratory to large scale production	2013	Applied Energy	453	2-s2.0-84871717568
Extinction risk and conservation of the world's sharks and rays	2014	eLife	450	2-s2.0-84898714955
Marine ecological processes	2016	Marine Ecological Processes	449	2-s2.0-84974777289

In addition, a randomised sample of 20 articles was reviewed. Nine of 20 articles do not closely reflect the theme of SDG 14. The majority of these articles are published in the field of chemistry and engineering. Others are related to historical climate change events and weather systems.

Sample of 20 randomised articles listed in Scopus from the updated (v3) string:

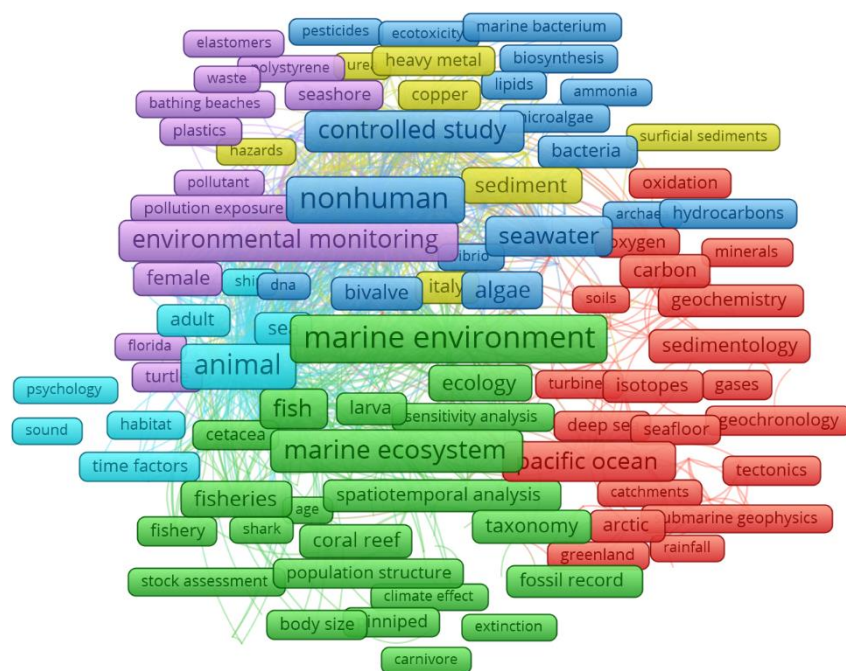
*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
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Calculation of Average Molecular Parameters, Functional Groups, and a Surrogate Molecule for Heavy Fuel Oils Using 1H and 13C Nuclear Magnetic Resonance Spectroscopy	2016	Energy and Fuels	23	2-s2.0-84971330284
Extraction of alginate biopolymer present in marine alga sargassum filipendula and bioadsorption of metallic ions	2013	Materials Research	18	2-s2.0-84876274950
Reconciling conflicts in pelagic fisheries under climate change	2015	Deep-Sea Research Part II: Topical Studies in Oceanography	17	2-s2.0-84924515916
Fluorescence, pigment and microscopic characterization of Bering Sea phytoplankton community structure and photosynthetic competency in the presence of a Cold Pool during summer	2014	Deep-Sea Research Part II: Topical Studies in Oceanography	12	2-s2.0-84909991491
Relative Impacts of Natural Stressors on Life History Traits Underlying Resilience of Intertidal Eelgrass (<i>Zostera marina</i> L.)	2013	Estuaries and Coasts	12	2-s2.0-84881550723
Evidence for seagrass meadows and their response to paleoenvironmental changes in the early Eocene (Jafnayn Formation, Wadi Bani Khalid, N Oman)	2016	Sedimentary Geology	11	2-s2.0-84975856163
Assessment of climate change downscaling and non-stationarity on the spatial pattern of a mangrove ecosystem in an arid coastal region of southern Iran	2016	Theoretical and Applied Climatology	10	2-s2.0-84988504497
Impact of typhoons on the composition of the upper troposphere within the Asian summer monsoon anticyclone: The SWOP campaign in Lhasa 2013	2017	Atmospheric Chemistry and Physics	5	2-s2.0-85017401008
Facial expression recognition using DCT features and neural network based decision tree	2013	Proceedings Elmar - International Symposium Electronics in Marine	5	2-s2.0-84892422435
Recent advances in marine N-cycle studies using 15N labeling methods	2016	Current Opinion in Biotechnology	4	2-s2.0-84969244157
A new salinity equation for sound speed instruments	2017	Limnology and Oceanography: Methods	3	2-s2.0-85029366222
Skin protective effects of <i>Nannochloropsis gaditana</i> extract on H2O2-stressed human dermal fibroblasts	2017	Frontiers in Marine Science	3	2-s2.0-85024407225
Analytical solution of the nitracline with the evolution of subsurface chlorophyll maximum in stratified water columns	2017	Biogeosciences	3	2-s2.0-85019203808
Improving efficiency of marine cycloidal propeller for coastal shipping	2015	RINA, Royal Institution of Naval Architects - International Conference on Ship and Offshore Technology, ICSOT India 2015: Coastal and Inland Shipping	1	2-s2.0-84988353949

A Statistical Model for Vessel-to-Vessel Distances to Evaluate Radar Interference	2017	Journal of Navigation	0	2-s2.0-85017499392
Automatic classification of pinnipeds based on their vocalizations and fusion of cepstral features	2016	3rd International Conference on Signal Processing and Integrated Networks, SPIN 2016	0	2-s2.0-84991728143
The Influence of Bionic Creatures and Natural Condition on Design Inspirtation	2016	Proceedings - 2015 IIAI 4th International Congress on Advanced Applied Informatics, IIAI-AAI 2015	0	2-s2.0-84964317544
Sediment monitoring to assess the environmental effect of relict sand dredging for beach nourishment	2015	Coastal and Beach Erosion: Processes, Adaptation Strategies and Environmental Impacts	0	2-s2.0-84955626079
Water content as a new tool for discrimination between some shellfishes	2015	Pakistan Journal of Biological Sciences	0	2-s2.0-84940380914
Technical considerations for conducting marine seismic, geotechnical and site surveys in arctic regions	2013	Society of Petroleum Engineers - SPE Arctic and Extreme Environments Conference and Exhibition, AEE 2013	0	2-s2.0-84897406176

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



The third update led to the addition of new topics to the document results. The top 20 most cited did not change but articles were found in the randomised sample that covered these new topics (e.g. an article on fisheries). The added keywords are also seen in the co-occurrence of keywords above, especially in relation to fisheries. The analyses do show that improvements are needed to enhance coverage of several targets (e.g. ecotourism, fishing rights) and to reduce the emphasis on chemistry – and engineering related articles.

The fourth update was used to place more emphasis on the targets related to societal (development, fishers) aspects and fisheries. It was also used to reduce the emphasis on the field of chemistry.

{local fisheries}	Added	Added to enhance coverage of target 14B
{artisanal fishers}	Added	Added to enhance coverage of target 14B
{fishing rights}	Added	Added to enhance coverage of target 14B
{species richness}	Added	Added to enhance coverage of target 14.2 and 14.4
{traditional ecological knowledge}	Added	Added to enhance coverage of target 14.7 and 14A
{small Island development states}	Added	Added to enhance coverage of target 14A
{marine quota}	Added	Added to enhance coverage of target 14.4
{marine economy}	Added	Added to enhance coverage of target 14B and 14C
{marine policy}	Added	Added to enhance coverage of target 14B and 14C
coast*	Added	Added to first combined query term to retrieve articles with hits on a combination of coast* with the other query terms, as it is one of the target habitats and is not clearly covered in the results
mangrove	Added	Added to first combined query term to retrieve articles with hits on a combination of mangrove with the other query terms, as it is one of the target habitats and is missing the current results

Updated query (4): 96,446 document results

TITLE-ABS-

KEY ((marine OR ocean OR oceans OR sea OR seas OR coast* OR mangrove) AND ({water cycle} OR {water cycles} OR {biogeochemical cycle} OR {biogeochemical cycles} OR {oceanic circulation model} OR {oceanic circulation models} OR {oceanic circulation modelling} OR {oceanic circulation modeling} OR {ice-ocean} OR eutrophicat* OR marine OR {coral bleach} OR {coral bleaching} OR {coastal management} OR {coastal habitat} OR {coastal habitats} OR {marine debris} OR {ocean acidification} OR (acidification AND seawater) OR {fishery} OR {fisheries} OR {overfishing} OR {sustainable yield} OR {marine protected area} OR {marine protected areas} OR {marine conservation} OR {ecotourism} OR {community based conservation} OR {community-based conservation} OR {marine land slide} OR {marine pollution} OR {nutrient runoff} OR {coastal ecotourism} OR {destructive fishing} OR {local fisheries} OR {artisanal fishers} OR {fisheries rights} OR {species richness} OR {traditional ecological knowledge} OR {small Island development states} OR {marine quota} OR {marine economy} OR {marine policy}) AND NOT ({paleoclimate} OR {paleoceanography} OR {radiocarbon} OR {genetics} OR {medicine} OR {drug} OR {engineering} OR {aerosol})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search returned 96,446 document results compared to 92,535 document results retrieved by query version 3.

The top 20 cited articles were reviewed. Three of the 20 articles are not closely related to the theme of the SDG. Specifically, these articles related to biochemistry and renewable resources.

Top 20 most cited articles listed in Scopus from the updated (v4) string:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Climate change 2013 the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	3536	2-s2.0-84926664150
Plastic waste inputs from land into the ocean	2015	Science	1264	2-s2.0-84923095173
Carbon and other biogeochemical cycles	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	924	2-s2.0-84928642948
Status and ecological effects of the world's largest carnivores	2014	Science	918	2-s2.0-84892185014
Impacts of ocean acidification on marine organisms: Quantifying sensitivities and interaction with warming	2013	Global Change Biology	835	2-s2.0-84877016612
Plastic Pollution in the World's Oceans: More than 5 Trillion Plastic Pieces Weighing over 250,000 Tons Afloat at Sea	2014	PLoS ONE	774	2-s2.0-84914176492
Toxicity, mechanism and health effects of some heavy metals	2014	Interdisciplinary Toxicology	661	2-s2.0-84910613290
Mercury as a global pollutant: Sources, pathways, and effects	2013	Environmental Science and Technology	625	2-s2.0-84878224971
Global imprint of climate change on marine life	2013	Nature Climate Change	611	2-s2.0-84886722508
Processes and patterns of oceanic nutrient limitation	2013	Nature Geoscience	593	2-s2.0-84883444382
Global warming and recurrent mass bleaching of corals	2017	Nature	581	2-s2.0-85015676975
Global conservation outcomes depend on marine protected areas with five key features	2014	Nature	581	2-s2.0-84893920384
Renewable energy resources: Current status, future prospects and their enabling technology	2014	Renewable and Sustainable Energy Reviews	532	2-s2.0-84905676737
Microplastic ingestion by zooplankton	2013	Environmental Science and Technology	511	2-s2.0-84879246992
Contribution of Antarctica to past and future sea-level rise	2016	Nature	500	2-s2.0-84962381525

Extended reconstructed sea surface temperature version 4 (ERSST.v4). Part I: Upgrades and intercomparisons	2015	Journal of Climate	488	2-s2.0-84923008551
Multiple stressors of ocean ecosystems in the 21st century: Projections with CMIP5 models	2013	Biogeosciences	471	2-s2.0-84885110424
Evaluation of the ECMWF ocean reanalysis system ORAS4	2013	Quarterly Journal of the Royal Meteorological Society	459	2-s2.0-84880703032
Biodiesel from microalgae: A critical evaluation from laboratory to large scale production	2013	Applied Energy	457	2-s2.0-84871717568
Extinction risk and conservation of the world's sharks and rays	2014	eLife	453	2-s2.0-84898714955

In addition, a randomised sample of 20 articles was reviewed. Six of the 20 articles are not closely related to the theme of the SDG. Except for one article, all relate to the marine environment or marine organisms, but not to the SDG as the articles covered topics on public health and engineering. One article was included on terrestrial ecosystems due to the name of mountain range as a study area (sea hills). The random sample shows a better focus on societal aspects of the SDG, e.g. marine governance and marine policy.

Sample of 20 randomised articles listed in Scopus from the updated (v4) string:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Fate of nano- and microplastic in freshwater systems: A modeling study	2017	Environmental Pollution	79	2-s2.0-85000671239
Poriferan chitin as a template for hydrothermal zirconia deposition	2013	Frontiers of Materials Science	41	2-s2.0-84881611819
Molecular data reveal cryptic lineages within the northeastern Atlantic and mediterranean small mussel drills of the <i>Ocenebrina edwardsii</i> complex (Mollusca: Gastropoda: Muricidae)	2013	Zoological Journal of the Linnean Society	22	2-s2.0-84886441938
Risk Factors for Bartonella species Infection in Blood Donors from Southeast Brazil	2016	PLoS Neglected Tropical Diseases	17	2-s2.0-84962343444
Tracing the origin of paralytic shellfish toxins in scallop <i>Patinopecten yessoensis</i> in the northern Yellow Sea	2013	Food Additives and Contaminants - Part A Chemistry, Analysis, Control, Exposure and Risk Assessment	14	2-s2.0-84886376938
Influence factors of fracability in nonmarine shale	2014	Society of Petroleum Engineers - European Unconventional Resources Conference and Exhibition 2014: Unlocking European Potential	11	2-s2.0-84905994818
Acoustic and anisotropic multi-parameter seismic full waveform inversion case studies	2014	SEG Technical Program Expanded Abstracts	10	2-s2.0-85012873156

Acacia trees on the cultural landscapes of the Red Sea Hills	2014	Biodiversity and Conservation	10	2-s2.0-84912031192
Early Cenozoic Differentiation of Polar Marine Faunas	2013	PLoS ONE	10	2-s2.0-84872433812
Recruitment failure of scleractinian corals in a subtropical marginal environment: Three-year monitoring in a Hong Kong marine park	2017	Marine Pollution Bulletin	3	2-s2.0-85020780454
Genome sequence of Arenibacter algicola strain TG409, a hydrocarbon-degrading bacterium associated with marine eukaryotic phytoplankton	2016	Genome Announcements	3	2-s2.0-85010425358
Phylum Nematomorpha	2014	Thorp and Covich's Freshwater Invertebrates: Ecology and General Biology: Fourth Edition	3	2-s2.0-84942912534
Actualizing marine policy engagement	2017	Dialogues in Human Geography	2	2-s2.0-85026340790
"Sharks in Your Hands"—A Case Study on Effects of Teaching Strategies to Change Knowledge and Attitudes Towards Sharks	2016	Journal of Biological Education	1	2-s2.0-84939494816
Data-driven modeling of coastal water quality using the Bayesian method for coastal management	2016	Journal of Coastal Research	0	2-s2.0-84987729403
Observations on the laboratory culture of the harpacticoid copepod Euterpina acutifrons (dana, 1847) using different diets	2016	Indian Journal of Fisheries	0	2-s2.0-85029653393
European Corrosion Congress, EUROCORN 2015	2015	European Corrosion Congress, EUROCORN 2015	0	2-s2.0-85025601224
Estimation of river discharge based on remote sensing of a river plume	2015	Proceedings of SPIE - The International Society for Optical Engineering	0	2-s2.0-84961843800
Sustainable governance in relation to the financial aspect in managing coastal areas: Malaysian experience	2015	Planning Malaysia	0	2-s2.0-84959467254
Slender marine pile design for gravity loads	2013	Ports 2013: Success Through Diversification - Proceedings of the 13th Triennial International Conference	0	2-s2.0-84887360471

Graphical summary of co-occurrence of terms from 2000 randomly selected results:

